Who are the star products?

Today's question

Which products bring most of the money?

groupby sorts the receipts into piles

IN DAILY LIFE

Receipts sorted into piles, then each pile added up.

IN EXCEL

A PivotTable.

THE LIMIT

The result has a row label; reset_index flattens it.

```
by_product = sales_clean.groupby("stock_code")["revenue"]
by_product = by_product.sum()
by_product = by_product.sort_values(ascending=False)
by_product.head()
```

A running total needs a sorted list first

IN DAILY LIFE

A running total on a receipt roll.

IN EXCEL

=SUM(B2:B2) dragged down.

THE LIMIT

Sort first, or the running total means nothing.

```
total = by_product["revenue"].sum()
by_product["cumulative"] = by_product["revenue"].cumsum()
by_product["share"] = by_product["cumulative"] / total
```

ABC puts products into groups

IN DAILY LIFE

A few items hold most money.

IN EXCEL

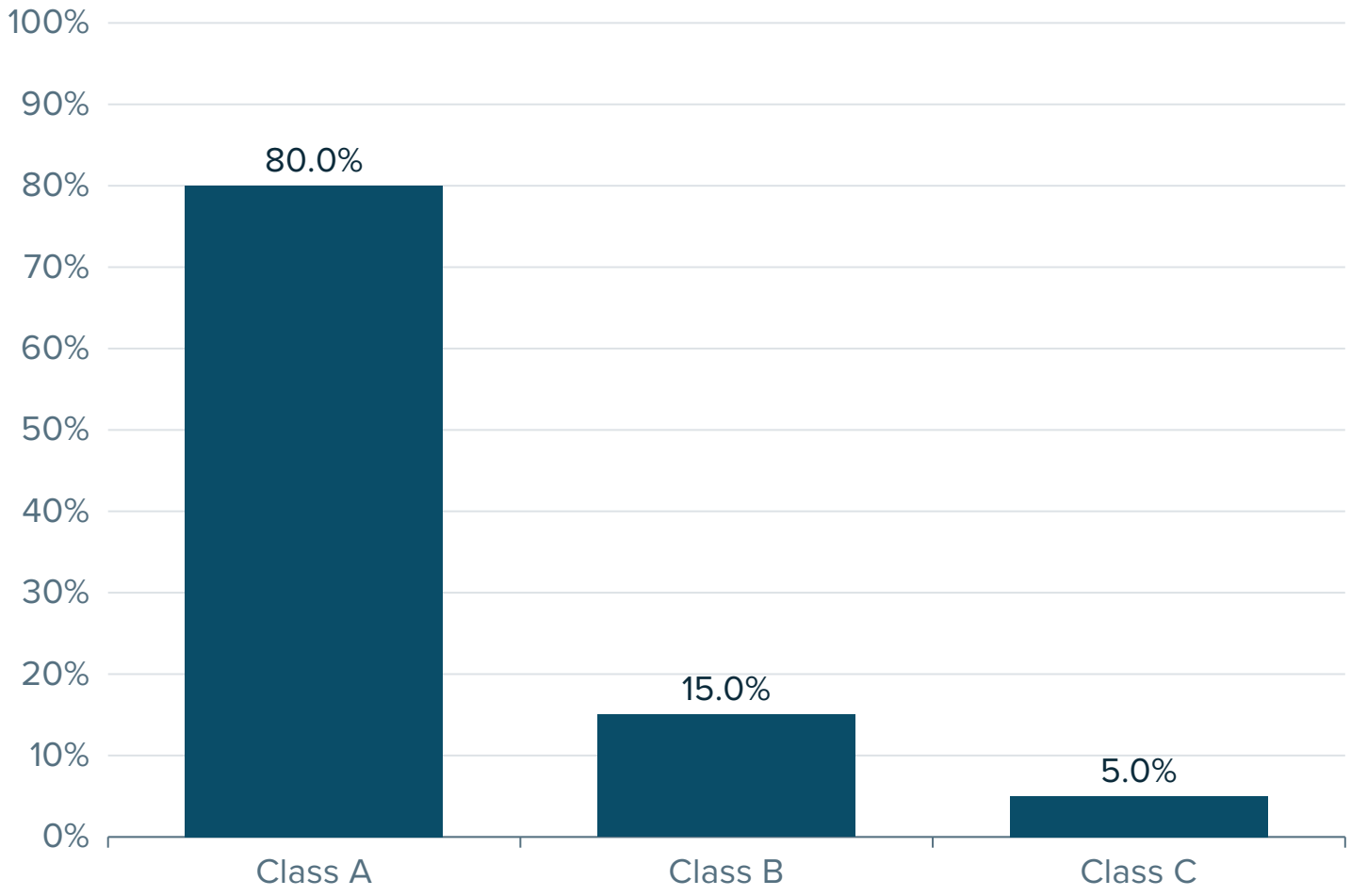
Sort, running %, IF.

THE LIMIT

A habit, not a law.

```
def abc_class(cumulative_share):
    if cumulative_share <= 0.80:
        return "A"
    elif cumulative_share <= 0.95:
        return "B"
    return "C"
```

About 1 in 5 products bring 80% of the money



80%

of revenue from class A

21.9%

of products are class A

Clean slice, Dec 2010 to Feb 2011

Now in Colab

Notebook	s5_star_products.ipynb
Cold test	20 minutes, notes allowed
After	One solution together

Three things you can do now

Rank products

groupby and sort

See the concentration

a running share

Classify

A, B, C with a function

Homework and daily recall

Homework

1. Finish the copy-alongs
2. Write your own ABC sentence

Daily recall

1. Three questions Tuesday
2. Three questions Wednesday